

The lattice is initialized at the base layer, $z = 0$, with a fraction P_0 of active sites. Each neighboring site is subsequently activated with probability $p(t = 0)$. The activation probability evolves according to

$$p(t + 1) = p(t) + c[f_T - f(t)], \quad (1)$$

where

$$f(t) = \frac{N(t)}{L^{d-1}}, \quad (2)$$

where d is the dimension in which the network is embedded.

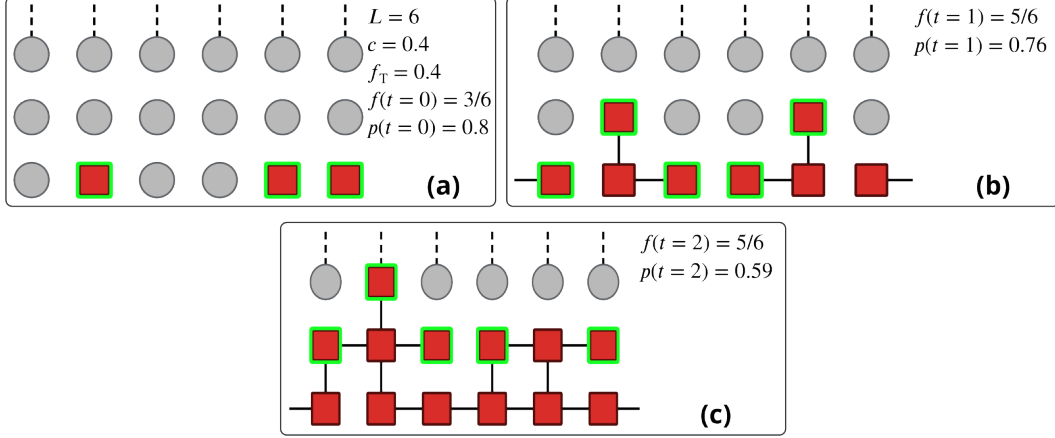


Figure 1: A lattice of side length $L = 6$, with feedback parameter $c = 0.4$, target fraction $f_T = 0.4$, initial fraction of active sites $f(t = 0) = 0.5$, and initial bond probability $p(t = 0) = 0.8$. Gray circles represent empty sites, red squares represent occupied sites, and red squares outlined in green indicate active occupied sites.

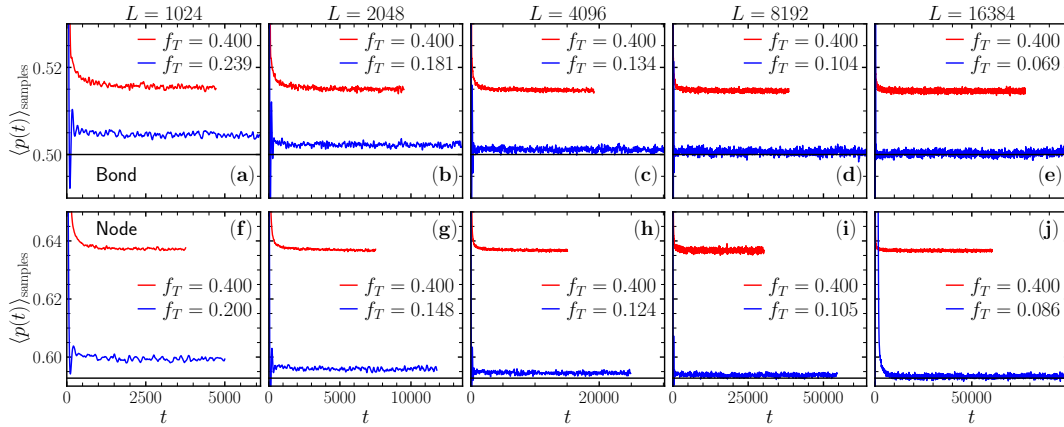


Figure 2: Sample-averaged occupation probability, $\langle p(t) \rangle$, as a function of the iteration time t . We considered $N_{\text{samples}} = [500, 400, 200, 100, 50]$ independent realizations for lattices of side length $L = [1024, 2048, 4096, 8192, 16384]$, respectively, with $c = 0.01$. The blue curve corresponds to the lowest value of f_T for which all realizations reached a stationary state, whereas the red curve corresponds to $f_T = 0.4$. The horizontal black lines indicate the critical threshold of conventional two-dimensional percolation for the corresponding percolation rule. **Upper panel:** Time series for bond percolation, for which the conventional critical threshold is $p_c = 0.5$ [?]. **Lower panel:** Time series for site percolation, for which the conventional critical threshold is $p_c = 0.59274(10)$ [?].

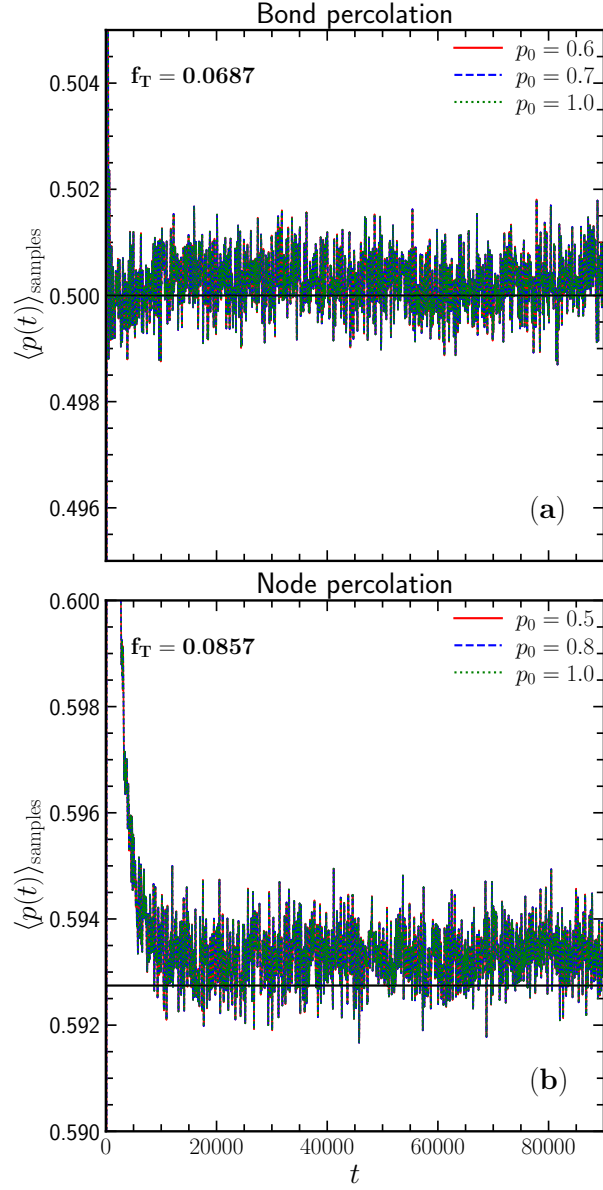


Figure 3: Sample-averaged occupation probability, $\langle p(t) \rangle$, as a function of the iteration time t . We considered $N_{\text{samples}} = 50$ independent realizations of a lattice with side length $L = 16384$ and feedback parameter $c = 0.01$. For each case, we selected the lowest value of f_T for which all realizations reached a stationary state. **Upper panel:** Bond-percolation process. The solid red, dashed blue, and dotted green curves correspond to the initial probabilities $p(t = 0) = p_0 = 0.6, 0.7$, and 1.0 , respectively. The horizontal black line indicates the critical threshold for conventional bond percolation, $p_c = 0.5$. **Lower panel:** Site-percolation process. The solid red, dashed blue, and dotted green curves correspond to $p_0 = 0.5, 0.8$, and 1.0 , respectively. The horizontal black line indicates the critical threshold for conventional site percolation, $p_c = 0.59274(10)$.

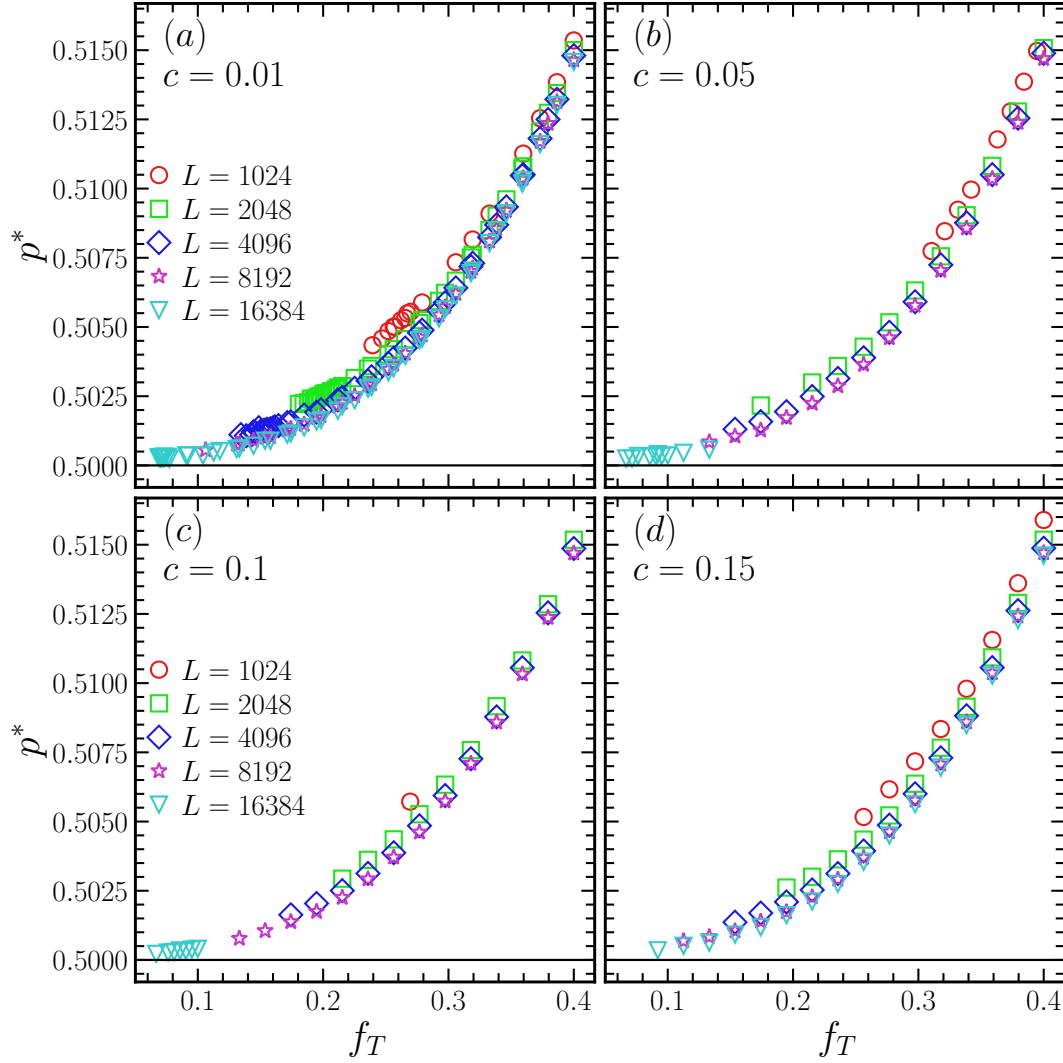


Figure 4: Mean stationary value of the activation probability $p(t)$, denoted by p^* , as a function of the target fraction f_T . The bond-percolation process was simulated using $N_{\text{samples}} = [500, 400, 200, 100, 50]$ independent realizations for lattices of side length $L = [1024, 2048, 4096, 8192, 16384]$, respectively. The feedback parameters $c = [0.01, 0.05, 0.10, 0.15]$ were used in panels (a)–(d), respectively. Red circles, green squares, blue diamonds, purple stars, and cyan triangles correspond to $L = 1024, 2048, 4096, 8192$, and 16384 , respectively. The horizontal black line indicates the critical threshold for conventional bond percolation, $p_c = 0.5$.

name

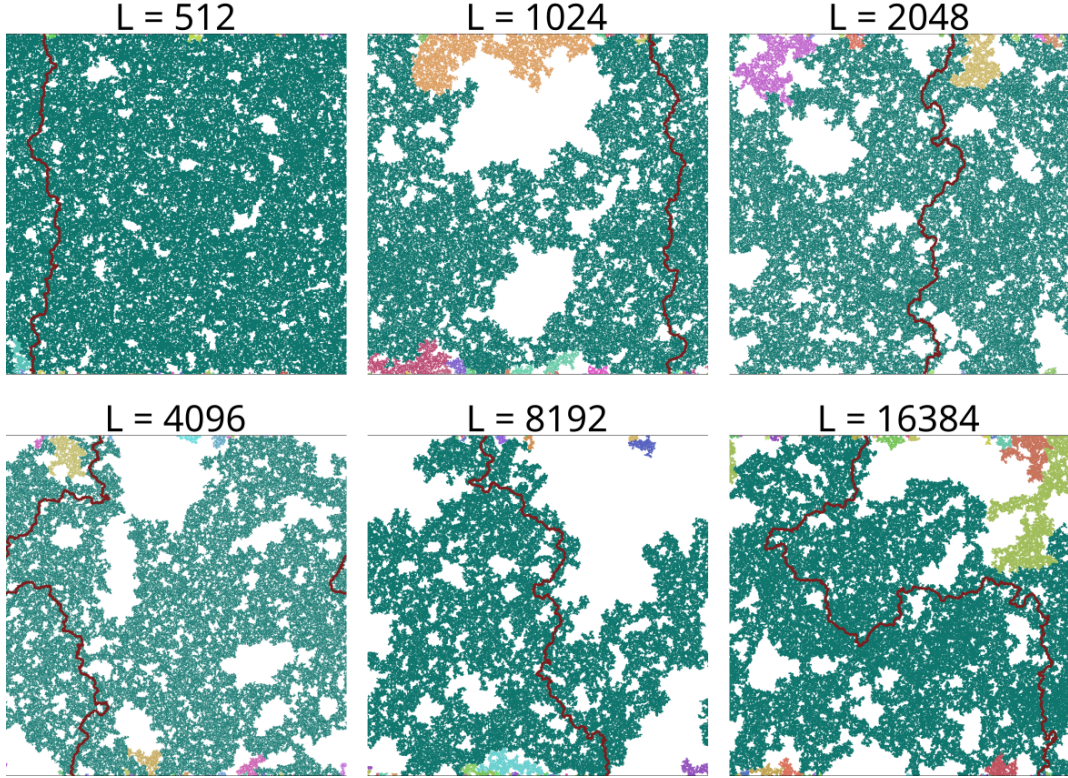


Figure 5: Sites connected by open bonds in the bond-percolation process. All lattices were generated using a feedback weight $c = 0.01$, side length L , an initial activation of $P_0 L = 0.2L$ sites at the base, and an initial activation probability $p(t = 0) = 0.8$. Dark-green sites represent the largest connected component, while the remaining colors identify the other clusters. The red path indicates the shortest path connecting the base to the top of the lattice.

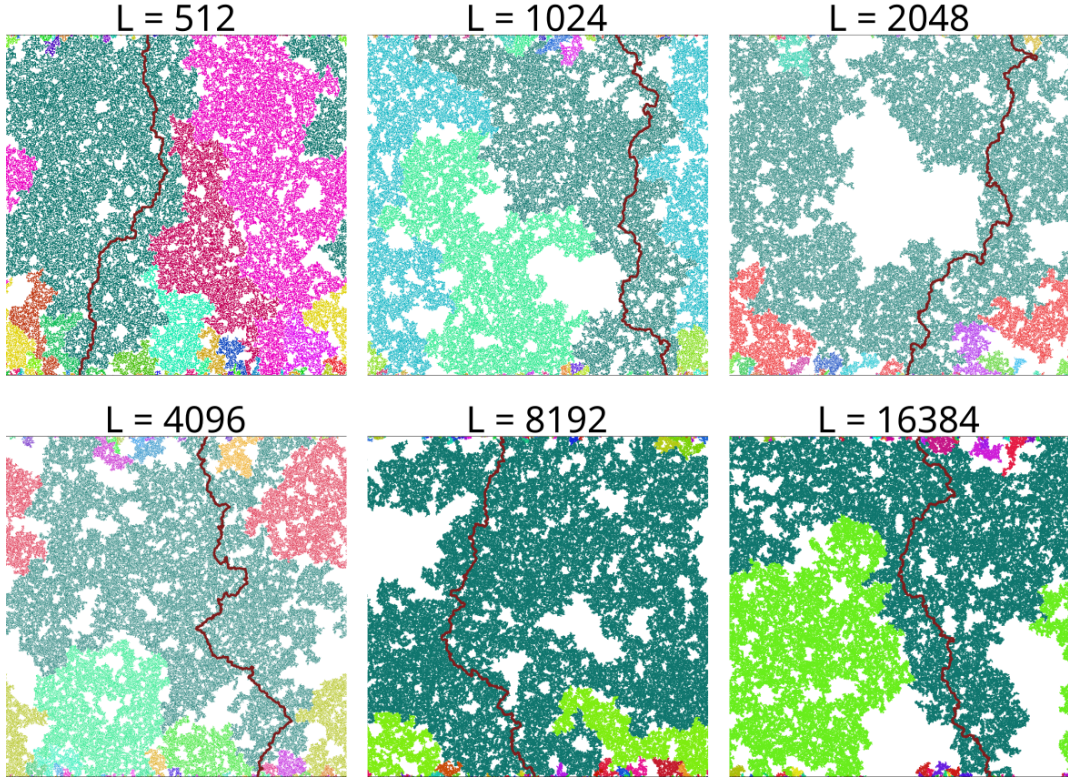


Figure 6: Connected sites in the bond-percolation process. All lattices were generated with feedback weight $c = 0.01$, side length L , $P_0 L = 0.2L$ initially active sites at the base, and initial activation probability $p(t = 0) = 0.8$. Dark-green sites represent the largest connected component, whereas the remaining colors distinguish the other clusters. The red curve indicates the shortest path connecting the base to the top of the lattice.

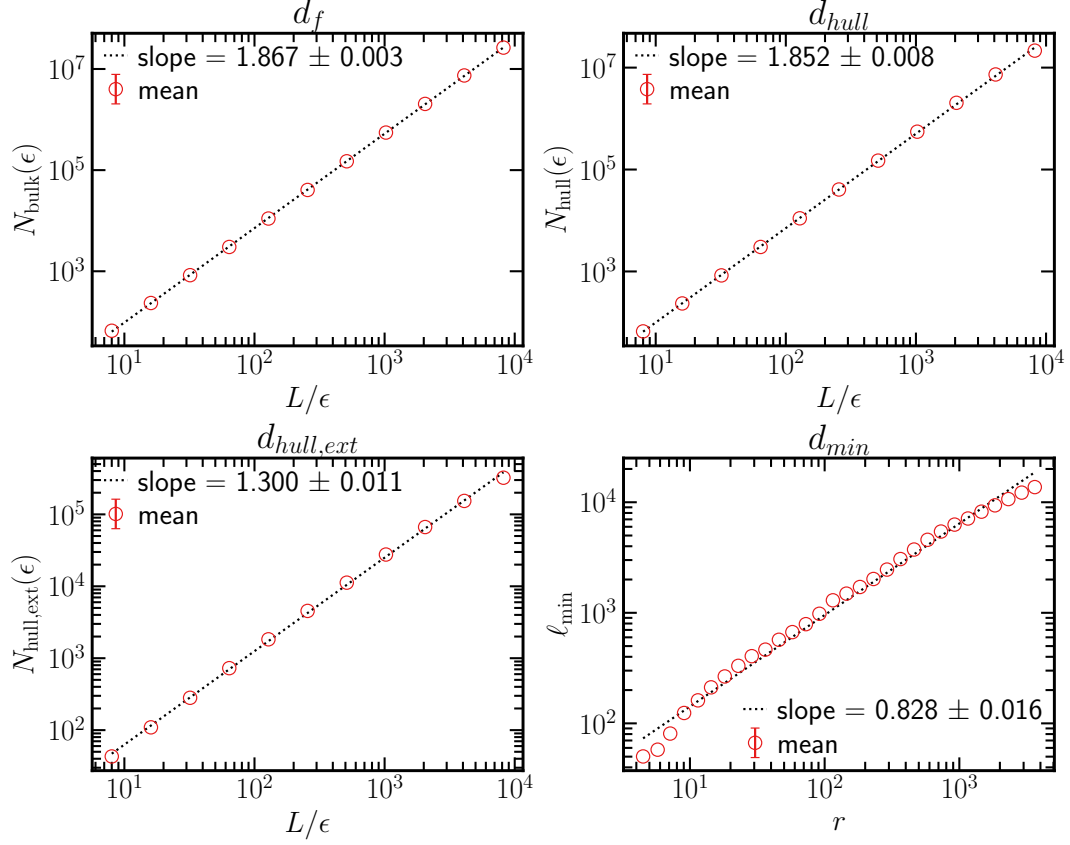


Figure 7: Propriedades topologicas calculadas para rede de comprimento de lado $L = 16384$ com peso do feedback $c = 0.01$, fração de ativação inicial $f(t = 0) = 0.2L$, probabilidade de ativação inicial $p_0 = 0.8$, e $f_T = f_{T,min} = 0.06873726$, sendo esse o menor valor obtido onde todas as amostras estabilizaram, sendo utilizadas $N_{\text{samples}} = 50$ amostras. Nos temos a dimensao fractal do bulk numa rede 2D quadrada regular dada por $d_f = 1.896$, $d_{\text{hull}} =$, $d_{\text{min}} = 1.130(2)$ [?], $d_{\text{hull,ext}} = 2$